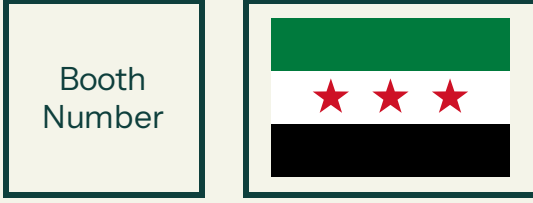


NASAQ: AI Task Orchestration and Aerial Data Analysis for Drone Swarms Under Human Command

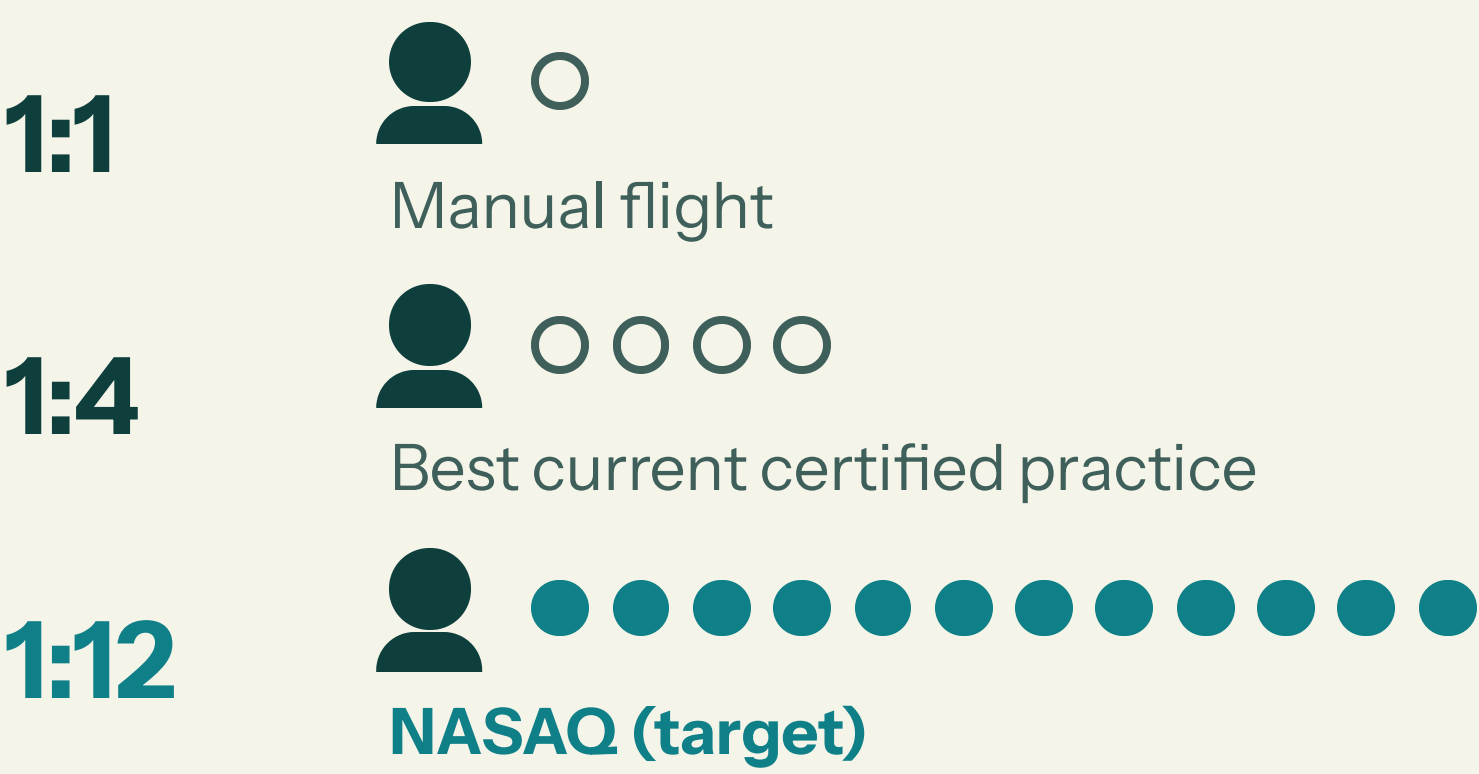
Student Name: MAHNOUR TAHER, AMR ALMASRI ID: _____

One operator sets the mission. NASAQ orchestrates the swarm and analyzes what it sees.



Introduction

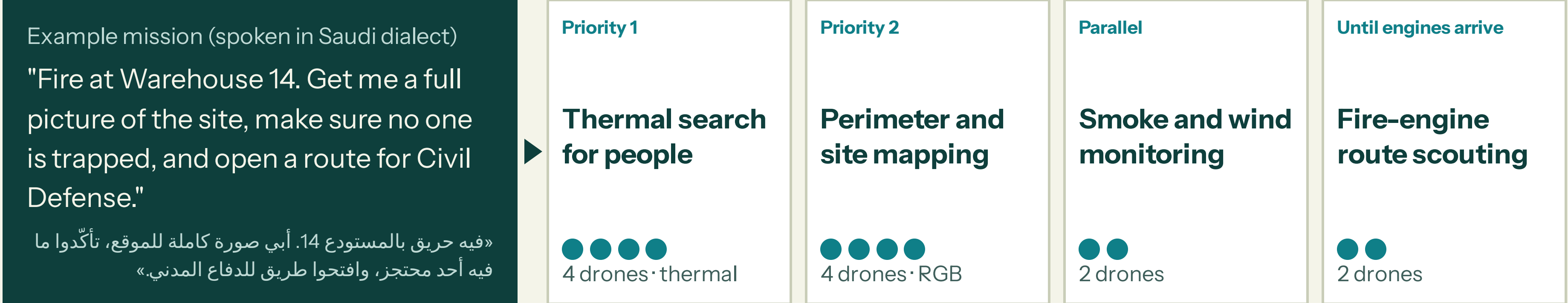
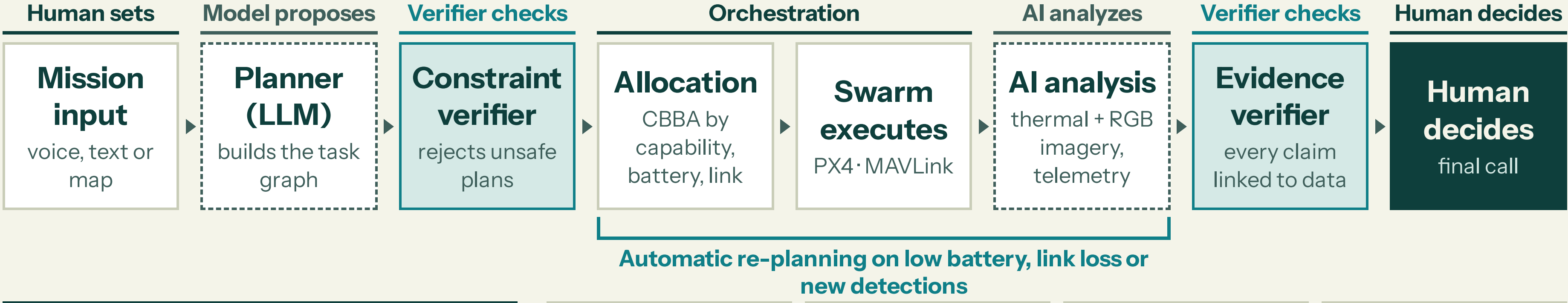
Drone swarms can cover a whole site, but today every drone is flown by hand and every image stream and alert lands on a human.



Operator performance drops once one person supervises more than about three robots [1], and LLM planners used alone still produce invalid task orders and infeasible actions [1]. Operations centers and Civil Defense need the coordination handled for them.

Goal: the human sets the mission and makes the key decisions; NASAQ orchestrates the tasks across the swarm and analyzes the drone data with AI.

Methodology

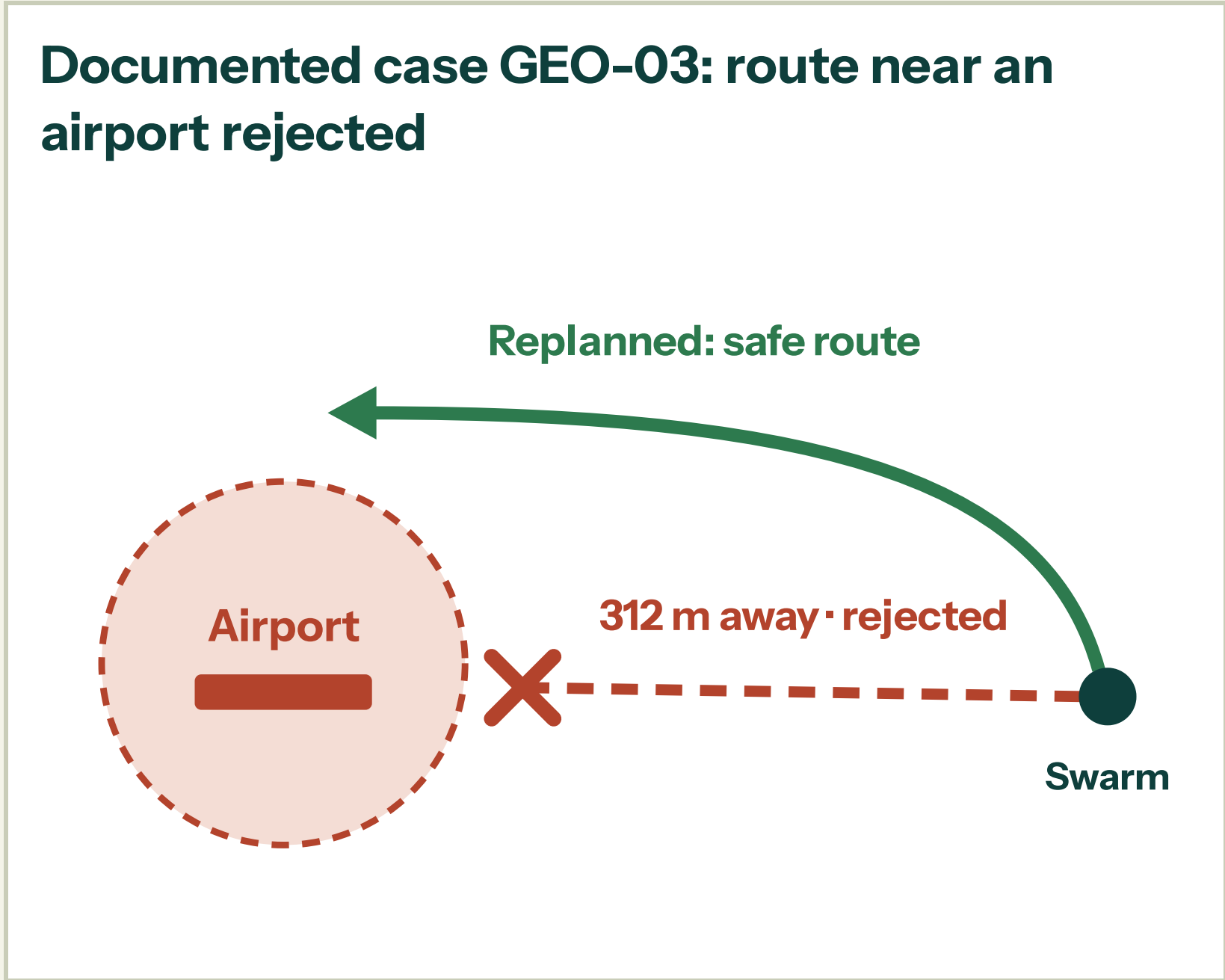


- Orchestration:** each mission is compiled into a task graph with dependencies and priorities; CBBA allocates tasks by sensor capability, battery and link quality, and re-allocates on events without operator input.
- AI analysis of drone data:** detection models analyze thermal and RGB imagery (hotspots, smoke, people as heat signatures only, vehicles), and a language model turns the findings into an explained alert.
- Deterministic safety:** a learning-free verifier aligned with ASTM F3269 checks every plan and re-allocation before and during flight. Tested with 12 drones in PX4 SITL and 3 to 4 real drones indoors.

Results

Measurement is ongoing under a pre-registered protocol, with final results before 1 November 2026. The "Measured" column will be filled with actual data only.

Metric	Target	Measured
Operator-to-drone ratio	1:12	Measuring
Decision time per alert	90 → 15 s	Measuring
Mission understanding: Fusha / dialects	≥90% / ≥85%	Measuring
Verifier: violations caught / false rejections	≥95% / ≤5%	Measuring
Explanation faithfulness (evidence-backed sentences)	100%	Measuring
False alarms per drone-hour	≤0.5	Measuring
Link loss followed by safe return	10/10	Measuring



- Declared limitations**
- AI person detection from altitude is validated in simulation only.
 - Spontaneous Saudi-dialect speech has about 37% word error rate, so mission text is confirmed before execution.
 - Coordination is centralized in the first version.
 - Models are used off the shelf; fine-tuning is planned (see Future Work).

Innovation

- Orchestration, not piloting:** one mission becomes a task graph for 12 drones, re-planned automatically as conditions change.
- Model proposes, verifier checks, human decides:** the language model sees only area names and task types and never touches a motor.
- AI reads what the swarm sees:** thermal and RGB analysis from two drones is fused into one alert, with every sentence linked to its evidence.
- Change intent in one sentence:** "Priority to the east" reorganizes the whole swarm without steering any single drone.

Example: an AI-explained alert awaiting a human decision

⚠ Possible fire · Warehouse 14

D1-0047

AI analysis: not an AC unit. Its heat keeps rising and never stabilizes, and a second drone confirmed it.

E-117 thermal

E-121 RGB smoke

38°C → 62°C

Attribution check 5/5

Awaiting your decision

Dispatch Civil Defense

Example from a pilot simulation.

Conclusion

NASAQ moves the operator from piloting drones to leading the mission: it compiles one mission into coordinated tasks for 12 drones, analyzes their imagery with AI, and brings each alert to a human with its evidence, targeting a 15-second decision instead of 90.

It runs on on-premise servers so no data leaves the site, works with any MAVLink drone, and includes no facial recognition.

GACAR Part 107 Saudi PDPL NCA controls SDAIA AI Ethics

Future Work & References

Next steps (planned)

- Fine-tune the planner:** LoRA on an open model (e.g. ALLaM) using mission → task-graph pairs generated from PX4 SITL [4, 5].
- Fine-tune the detector:** FLAME 3 and HIT-UAV, then local rooftop and AC-unit footage [6, 7].
- Sensor integration:** site smoke and heat detectors, CCTV and wind data as mission triggers and extra evidence.
- 90-day field pilot with Civil Defense; decentralized coordination.

References

- J. Chen et al., "Melding LLM and temporal logic for reliable human-swarm collaboration in complex scenarios," arXiv:2605.07877, 2026.
- S. S. Kannan, V. L. N. Venkatesh, B.-C. Min, "SMART-LLM: Smart multi-agent robot task planning using large language models," IROS 2024, arXiv:2309.10062.
- K. Liu et al., "COHERENT: Collaboration of heterogeneous multi-robot system with large language models," ICRA 2025, arXiv:2409.15146.
- A. Ahmed et al., "LLM-guided distributed model predictive control for decentralized UAV formations," IEEE Access, 2025, doi:10.1109/ACCESS.2025.3642325.
- I. Sharifi, A. Zongo, P. Wei, "Fine-tuning large language models for cooperative tactical deconfliction of small unmanned aerial systems," arXiv:2603.28561, 2026.
- B. Hopkins et al., "FLAME 3 dataset: Radiometric thermal UAV imagery for wildfire management," arXiv:2412.02831, 2024.
- J. Suo et al., "HIT-UAV: A high-altitude infrared thermal dataset for UAV-based object detection," Scientific Data 10, 227, 2023.
- H.-L. Choi, L. Brunet, J. P. How, "Consensus-based decentralized auctions for robust task allocation," IEEE T-RO 25(4), 912-926, 2009; ASTM F3269-21.

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